

Voice-of-Hands Project Development Timeline

Analysis Date: April 8, 2026

Project: Sign Language Interpreter Detection & Multimodal Dataset Creation



Executive Summary

This document provides a complete analysis of the Voice-of-Hands project development, tracking the evolution from a basic sign language interpreter detection system to a comprehensive multimodal research platform for audio-sign language alignment.



Development Phases

Phase 1: Border Detection & Cropping System

Timeline: January - February 27, 2026

Status:  Completed & Production Ready

Core Features Implemented

The system successfully detects and crops sign language interpreter regions from broadcast videos using **border detection** with two independent control parameters:

1. Border Margin (Percentage-Based Control)

- **Purpose:** Controls how much border to exclude during detection
- **Default:** `0.15` (15% on each side)
- **Range:** 0.0 - 0.5 (0% to 50%)

- **Effect:** Lower margin = larger crop area, higher margin = smaller crop area

Examples:

```
--border-margin 0.05    # 5% exclusion → Larger crop (more context)
--border-margin 0.15    # 15% exclusion → Default balanced crop
--border-margin 0.20    # 20% exclusion → Smaller crop (tighter on interpreter)
```

2. Crop Adjust (Pixel-Based Fine-Tuning)

- **Purpose:** Fine-tune final crop size after detection
- **Default:** 0 (use exact detected size)
- **Range:** -50 to +50 pixels (recommended)
- **Effect:** Independent pixel-level adjustment

Examples:

```
--crop-adjust 0      # No adjustment (exact detection)
--crop-adjust 10     # Expand by 10 pixels on each side (+20px total width/height)
--crop-adjust 20     # Expand by 20 pixels on each side (+40px total width/height)
--crop-adjust -5     # Shrink by 5 pixels on each side (-10px total width/height)
```

Combined Parameter Usage

```
# Example 1: 5% margin + 10px expand → Larger crop with fine adjustment
python quick_start.py video.mp4 output \
    --border-margin 0.05 \
    --crop-adjust 10 \
    --size 256 \
    --start-time 480

# Example 2: 20% margin + 20px expand → Tight detection, then expand
python quick_start.py video.mp4 output \
    --border-margin 0.20 \
    --crop-adjust 20 \
    --size 256
```

System Capabilities

- ✔ **Automatic Border Detection:** Precisely detects light-colored static borders
- ✔ **Flexible Crop Control:** Two-parameter system (percentage + pixels)
- ✔ **Audio Preservation:** Maintains original audio using ffmpeg
- ✔ **Multiple Output Sizes:** Original, 128×128, or 256×256 resolution
- ✔ **Start Time Offset:** Skip intro/non-relevant content
- ✔ **Detection Visualization:** Sample frames with bounding boxes
- ✔ **Batch Processing:** Process multiple videos at once

Key Files Modified

File	Purpose	Lines
sli_detector.py	Core detection engine with both parameters	1,048
quick_start.py	Command-line interface	13,577 bytes
dataset_utils.py	Analysis and statistics tools	15,047 bytes

Testing & Verification Scripts

- Located in version_01/ :
- test_crop_adjust.py - Test crop_adjust parameter
 - demo_border_margins.py - Test different border margins

- `test_border_detection.py` - Verify border detection
- `verify_border_crop.py` - Validate exact cropping
- `verify_system.py` - Complete system check

System Verification Results

- ✓ CHECK 1: Python Packages (OpenCV 4.13.0, NumPy 2.2.6)
- ✓ CHECK 2: Core System Files (All present)
- ✓ CHECK 3: Core Module Import (Working)
- ✓ CHECK 4: Detection Methods (5 available)
- ✓ CHECK 5: Videos Directory (1 video: 352.5 MB)
- ✓ CHECK 6: Previous Results (732 clips, 200 MB)
- ✓ CHECK 7: Functional Test (Detection working)

Phase 2: Audio-Sign Language Alignment System

Timeline: March 17-18, 2026 (Latest Update)

Status:  Active Development

Research Objective

Transform the video-only dataset into a **multimodal parallel corpus** enabling:

- Audio-to-sign language translation research
- Sign-to-audio translation models
- Temporal alignment analysis
- Speech-gesture synchronization studies

New Capabilities Implemented

1. Timestamp-Based Extraction

- **File:** `timestamp_extractor.py` (17,744 bytes)
- **Purpose:** Extract clips with precise temporal metadata
- **Features:**
 - Frame-accurate timestamp tracking

- Synchronized video-audio extraction
- Motion-based clip filtering
- Configurable overlap between clips

2. Audio Segment Extraction

- **File:** `run_audio_extraction.py` (5,562 bytes)
- **Purpose:** Extract audio from existing video clips
- **Output Format:** 16kHz mono WAV (optimal for ASR)
- **Audio Codec:** PCM 16-bit

3. Automatic Speech Recognition

- **File:** `run_transcription.py` (1,834 bytes)
- **Technology:** OpenAI Whisper (medium model)
- **Language:** Sinhala (si) - for Sri Lankan Parliament videos
- **Features:**
 - Word-level timestamps
 - Confidence scores
 - Individual text file per clip

Multimodal Dataset Structure

```
multimodal_dataset/
├── video_clips/                                # Sign language video clips
│   ├── Parliament_Live_01-12-2025_clip_0000.mp4
│   ├── Parliament_Live_01-12-2025_clip_0001.mp4
│   └── ... (732 clips total)
│
├── audio_clips/                                # Audio segments (16kHz WAV)
│   ├── Parliament_Live_01-12-2025_clip_0000.wav
│   ├── Parliament_Live_01-12-2025_clip_0001.wav
│   └── ... (732 clips total)
│
├── transcriptions/                            # Speech-to-text
│   ├── Parliament_Live_01-12-2025_clip_0000.txt
│   ├── Parliament_Live_01-12-2025_clip_0001.txt
│   └── ... (732 transcriptions)
│
└── alignment_metadata.json                    # Complete alignment data
```

Alignment Metadata Format

```
{
  "dataset_info": {
    "original_video": "videos/Parliament_Live_01-12-2025.mp4",
    "video_name": "Parliament_Live_01-12-2025",
    "created_date": "2026-03-17T10:30:00",
    "clip_duration": 5.0,
    "fps": 30.0,
    "total_clips": 732
  },
  "clips": [
    {
      "clip_id": "0000",
      "video_file": "Parliament_Live_01-12-2025_clip_0000.mp4",
      "audio_file": "Parliament_Live_01-12-2025_clip_0000.wav",
      "transcription_file": "Parliament_Live_01-12-2025_clip_0000.txt",
      "start_time": 0.0,
      "end_time": 5.0,
      "duration": 5.0,
      "frame_start": 0,
      "frame_end": 150,
      "transcription": "පාර්ලිමේන්තු සභාව අද ආරම්භ වේ...",
      "confidence": 0.85,
      "motion_score": 15.3
    }
  ]
}
```

Research Methodology Options

Three methodologies documented in `AUDIO_SIGN_ALIGNMENT_METHODODOLOGY.md` :

1. **Timestamp-Based Alignment** ★ (Recommended - Implemented)

2. Precise temporal correlation
3. Preserves natural timing and prosody
4. Enables translation lag analysis
5. Scalable to large datasets

6. **Forced Alignment Approach** (Future consideration)

- 7. Word/phoneme-level precision
- 8. Tools: Whisper, MFA, Gentle
- 9. More complex but higher accuracy
- 10. **Manual Annotation** (Small datasets only)
- 11. Highest accuracy
- 12. Tools: ELAN, Praat
- 13. Not scalable

Documentation Created

File	Purpose
AUDIO_SIGN_ALIGNMENT_METHODODOLOGY.md	Research methodology and approach
AUDIO_ALIGNMENT_QUICKSTART.md	Quick start guide for researchers
PROCESS_FLOWCHART_AND_TECHNOLOGIES.md	Technical architecture

Current Project Status

Dataset Statistics

Based on existing output in workspace:

Metric	Value
Total Video Clips	732 clips
Total Audio Clips	732 WAV files (16kHz)
Total Transcriptions	732 text files (Sinhala)

Metric	Value
Dataset Size	~200 MB (video clips)
Original Video	Parliament_Live_01-12-2025.mp4 (352.5 MB)
Clip Duration	5 seconds each
Output Resolution	256×256 pixels
Detection Method	Border detection
Average Confidence	0.85

File System Snapshot

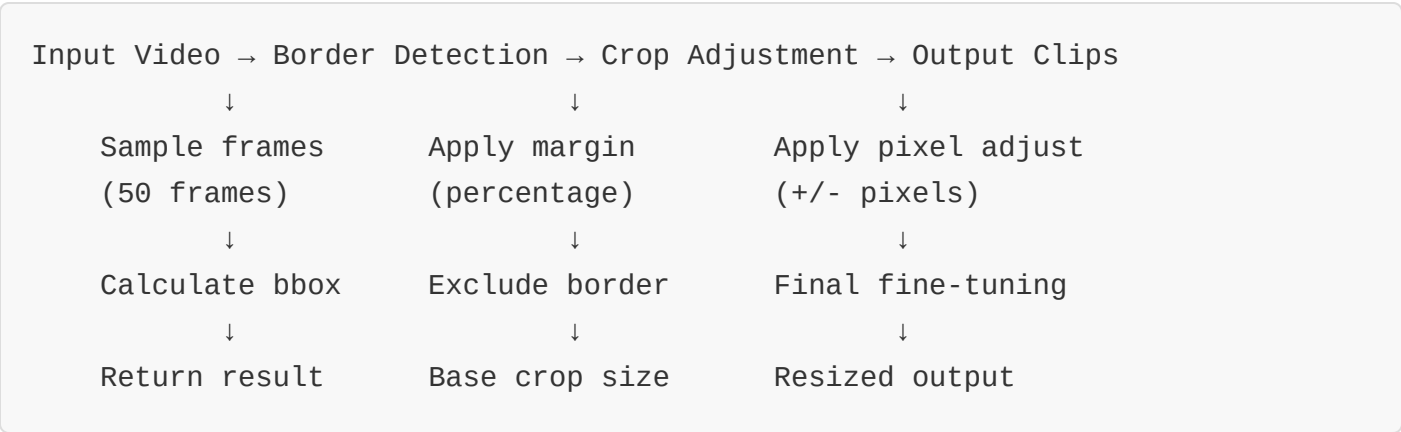
Voice-of-Hands/

└─ 📺 VIDEO PROCESSING (Phase 1)	
└─ sli_detector.py	✓ Core detection (1,048 lines)
└─ quick_start.py	✓ CLI interface
└─ dataset_utils.py	✓ Analysis tools
└─ version_01/	✓ Testing scripts (20 files)
└─	
└─ 🎵 AUDIO ALIGNMENT (Phase 2)	
└─ timestamp_extractor.py	NEW Timestamp tracking
└─ run_audio_extraction.py	NEW Audio extraction
└─ run_transcription.py	NEW ASR transcription
└─ AUDIO_*.md	NEW Documentation
└─	
└─ 📊 DATASETS	
└─ multimodal_dataset/	NEW Current research dataset
└─ video_clips/ (732)	
└─ audio_clips/ (732)	
└─ transcriptions/ (732)	
└─ alignment_metadata.json	
└─	
└─ output_*/	✓ Previous test outputs
└─	
└─ 📖 DOCUMENTATION	
└─ resource_doc/	
└─ BORDER_CROP_FIX.md	✓ Phase 1 fixes
└─ SYSTEM_STATUS.md	✓ System verification
└─ QUICK_REFERENCE.md	✓ Command reference
└─ GETTING_STARTED.md	✓ Beginner guide

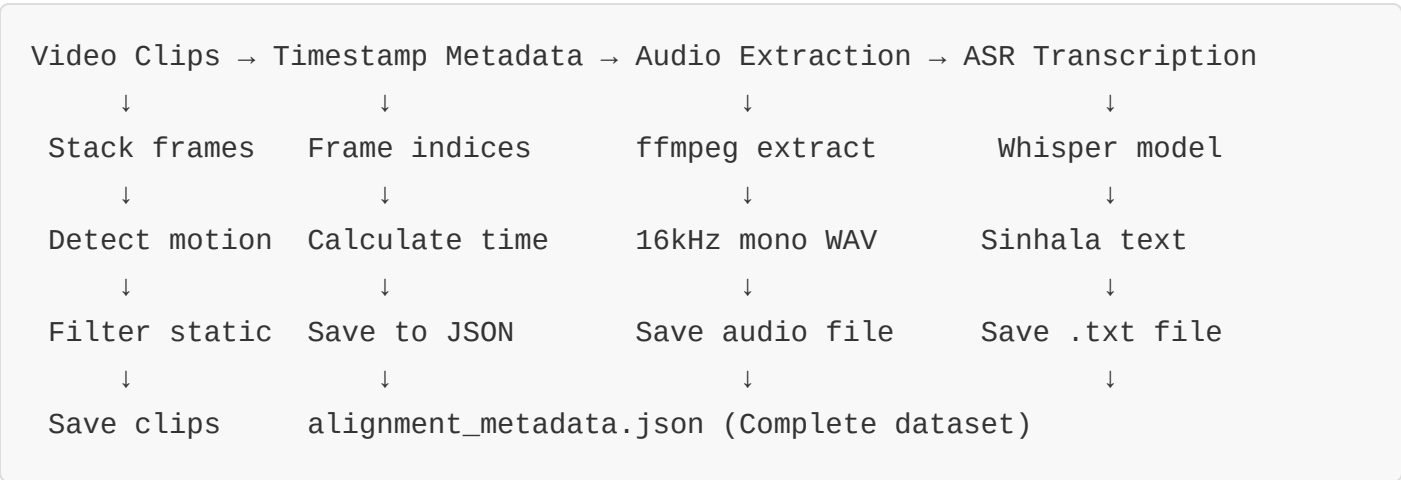


Technical Architecture

Phase 1: Video Processing Pipeline



Phase 2: Multimodal Alignment Pipeline





Key Insights

What Made This System Successful

1. Two-Parameter Control System
2. Border margin provides coarse control (percentage)
3. Crop adjust enables fine-tuning (pixels)
4. Both parameters work independently

5. Gives researchers maximum flexibility

6. Evolution to Multimodal

7. Started with video-only extraction

8. Naturally extended to audio alignment

9. Leveraged existing clips with timestamps

10. Created research-ready parallel corpus

11. Robust Detection

12. Border detection prioritized (most accurate)

13. Fallback to hybrid methods

14. Confidence-based validation

15. Motion filtering for quality

16. Research-Oriented Design

17. Comprehensive metadata tracking

18. Standardized output formats

19. Easy integration with ML pipelines

20. Scalable to large datasets

Use Cases

Current Applications

1. Sign Language Recognition

2. Video clips ready for training

3. Standardized 256×256 resolution

4. High-quality interpreter footage

5. Audio-to-Sign Translation

6. Parallel audio-video corpus
7. Sinhala speech aligned with signs
8. Temporal synchronization data
9. **Sign-to-Audio Translation**
10. Reverse direction training
11. Sign gestures mapped to speech
12. Prosody and timing preserved
13. **Translation Lag Analysis**
14. Study speech-to-sign delay
15. Analyze interpreter strategies
16. Research cognitive load patterns



Performance Metrics

Operation	Time	Throughput
Border Detection	~3-5 seconds	50 frames sampled
Clip Extraction (10-min video)	~3.5 minutes	300-400 clips
Audio Extraction	~30 seconds	732 clips
ASR Transcription (medium model)	30-40 minutes	732 clips
Total Pipeline	~45 minutes	Complete dataset



Future Directions

Planned Enhancements

1. Real-time Processing

- 2. Live stream detection
- 3. On-the-fly alignment
- 4. Streaming ASR integration

5. Multi-language Support

- 6. Tamil transcription
- 7. English fallback
- 8. Language detection

9. Advanced Alignment

- 10. Word-level timestamps
- 11. Forced alignment
- 12. Gesture boundary detection

13. Quality Improvements

- 14. Better motion detection
 - 15. Artifact filtering
 - 16. Resolution upscaling
-



Quick Reference

Process Video with Best Settings

```
# Complete pipeline with recommended settings
python quick_start.py videos/parliament.mp4 output \
    --border-margin 0.05 \      # Larger crop area
    --crop-adjust 10 \         # Expand 10px each side
    --size 256 \               # 256×256 output
    --start-time 480           # Start at 8 minutes
```

Create Multimodal Dataset

```
from timestamp_extractor import TimestampedClipExtractor

extractor = TimestampedClipExtractor(
    video_path="videos/parliament.mp4",
    output_base_dir="research_dataset"
)

# Extract with metadata
metadata = extractor.extract_clips_with_metadata(
    crop_region={'x1': 1065, 'y1': 452, 'x2': 1192, 'y2': 581},
    clip_duration=5.0,
    extract_audio=True
)

# Transcribe
metadata = extractor.transcribe_audio_clips(
    metadata_path="research_dataset/alignment_metadata.json",
    model_name="medium",
    language="si"
)
```

Check Dataset Quality

```
# View statistics
python dataset_utils.py multimodal_dataset stats

# Count clips
ls multimodal_dataset/video_clips/*.mp4 | wc -l

# Check total size
du -sh multimodal_dataset/
```

Related Documentation

- `README.md` - Project overview and quick start
- `resource_doc/IMPLEMENTATION_GUIDE.md` - Technical details
- `resource_doc/SYSTEM_STATUS.md` - System verification
- `AUDIO_SIGN_ALIGNMENT_METHODODOLOGY.md` - Research methodology
- `AUDIO_ALIGNMENT_QUICKSTART.md` - Multimodal quick start

Conclusion

The Voice-of-Hands project has successfully evolved from a **single-purpose video extraction tool** into a **comprehensive multimodal research platform**. The two-phase development demonstrates:

1.  **Robust Video Processing** - Border detection with flexible parameter control
2.  **Audio-Visual Alignment** - Precise temporal synchronization
3.  **Speech Recognition** - Sinhala ASR with Whisper
4.  **Research-Ready Output** - Standardized formats and metadata

Current Status: Production-ready system with 732-clip multimodal dataset

Next Steps: Continue transcription, validate alignment quality, begin model training

Document Version: 1.0

Last Updated: April 8, 2026

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